



Reflex Race

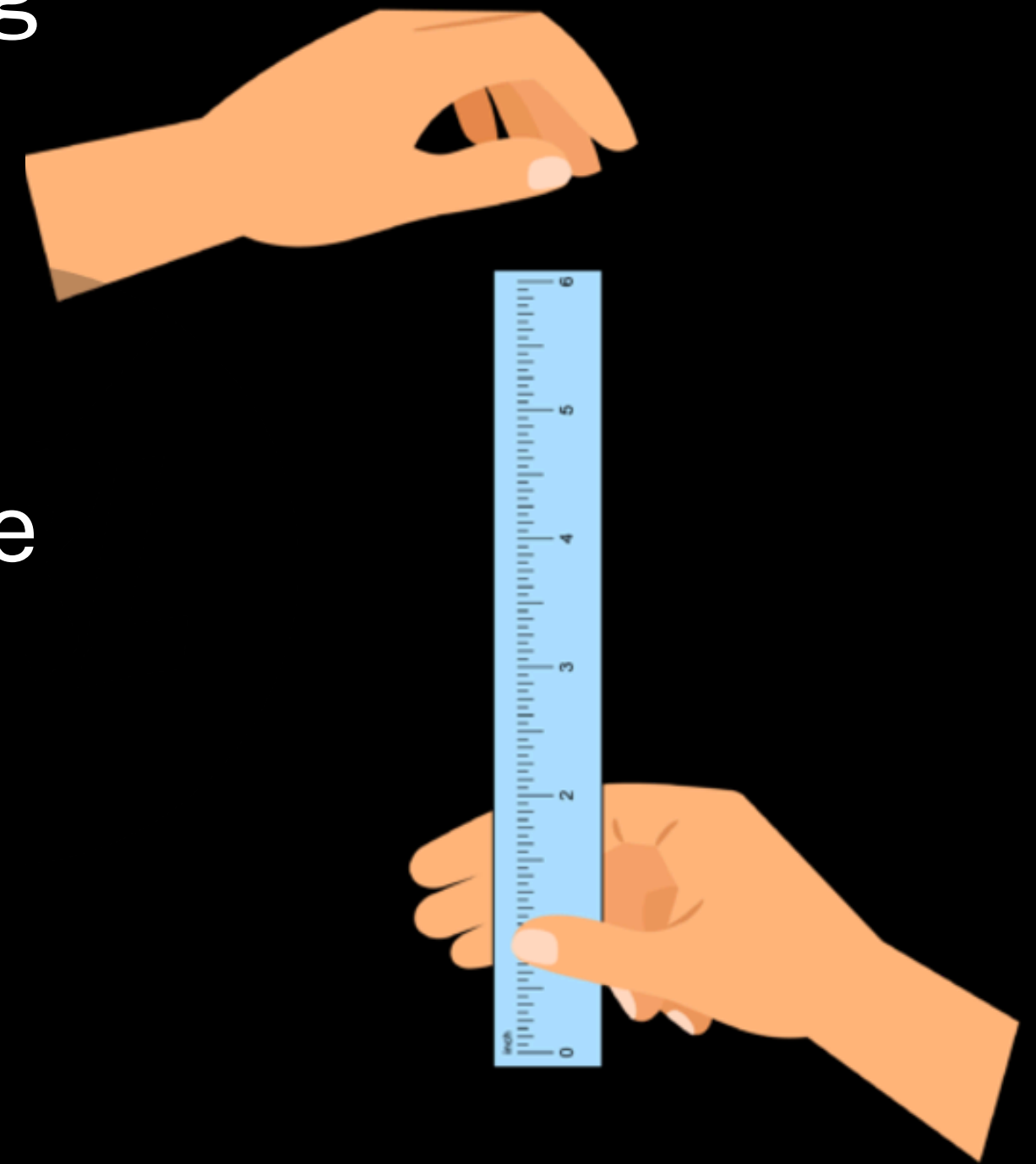
compete. react. win.

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The Problem

- Reflex testing is often static and unengaging
- No real-time competition or interaction
- Lack of dynamic feedback (speed/intensity)
- Existing setups do not simulate real pressure situations

→ **Need for an engaging, real-time reflex testing system**



The Solution

Possible approaches

Software-based reaction test

- Screen + keyboard input
- Arduino minimally used

Hardware interactive system

- LEDs + push buttons
- Timing + feedback control

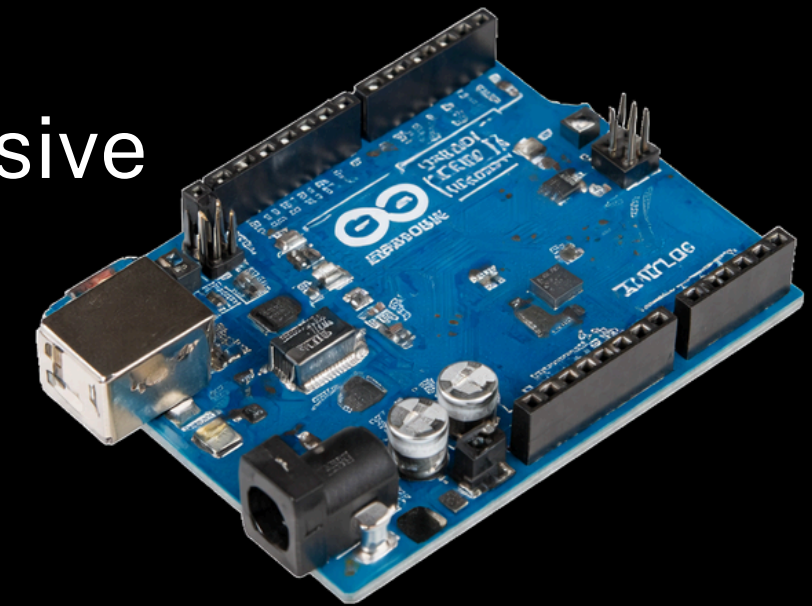
Comparision

Software:

- Simple but less immersive

Hardware:

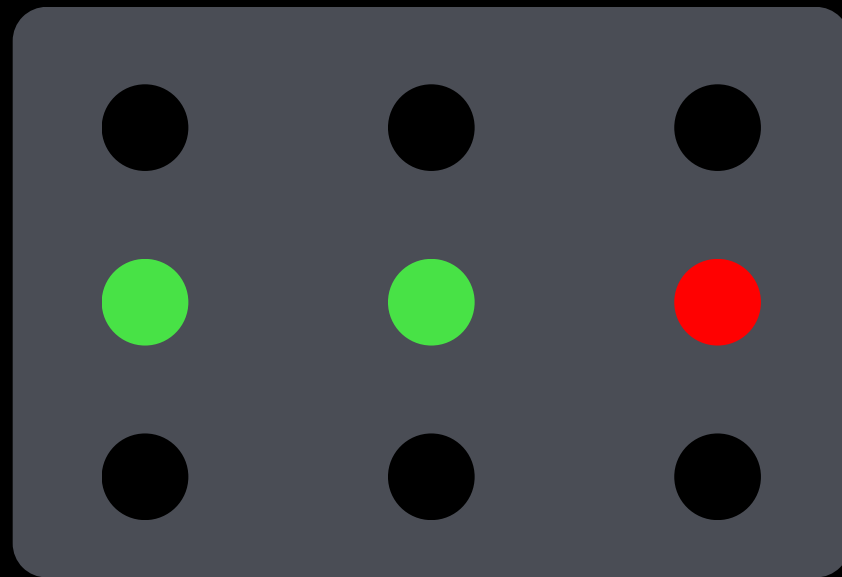
- Interactive and engaging
- Real-time feedback



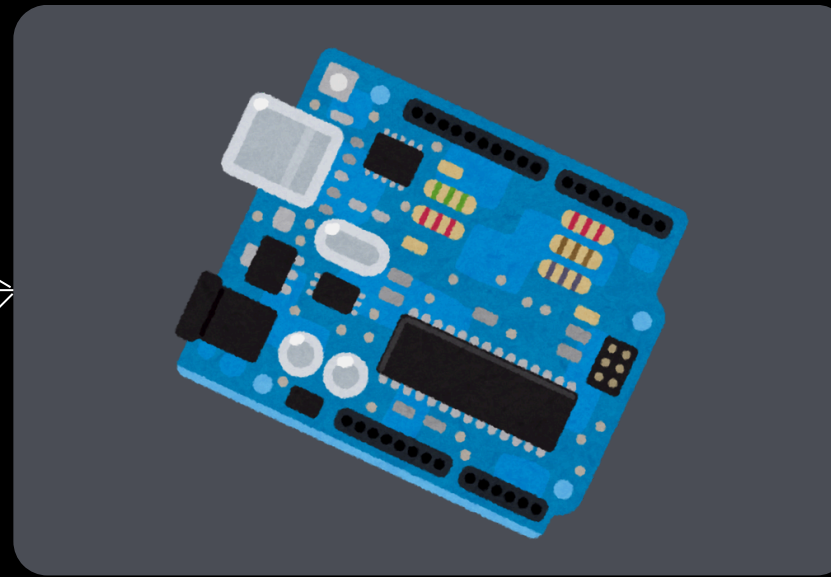
Chosen : Hardware Interactive System. Why?

- Real-time response
- Competitive gameplay
- Uses Arduino I/O + timing

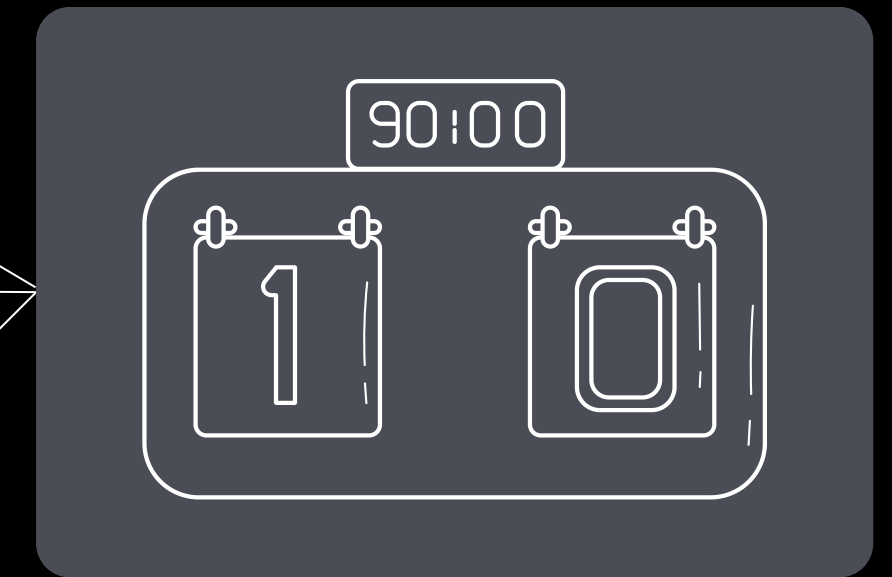
Solution Design



LED + Push buttons
(Input)



Arduino
(processing)



OLED Display + Buzzer
(output)

Solution Design

Could this be done without Arduino?

- Possible using discrete circuits or a computer-based system
- However, it would lack flexibility and real-time control

Role of Arduino :

- Handles precise timing and input processing
- Controls LEDs, buzzer, and display simultaneously
- Enables dynamic gameplay (speed variation, scoring)

Detailed Design

Component	Quantity
Arduino UNO	1
RGB LEDs	3
220 ohm Resistors	6
Piezo Buzzer	1
OLED Display	1
Push buttons	7

Libraries Used:

- Wire.h
- Adafruit SSD1306
- Adafruit GFX

Memory Optimization:

- F() macro used to store strings in program memory
- Reduces RAM usage on Arduino

What took so long?

- **Limited I/O pins on Arduino Uno**

- Optimized pin usage (used only RG in RGB LEDs, grounded Blue)

- **Display constraints**

- Switched from TFT to OLED (fewer pins, I2C communication)

- **Memory limitations**

- Optimized usage using F() macro

- **Non-blocking timing & synchronization**

- Ensured music, display, and game actions run without delays

Discussion of Results

- Real-time synchronization of LEDs, buzzer, and display
- Increasing difficulty with time makes gameplay more challenging
- OLED provides continuous feedback (timer, scores, reaction time)
- Competitive 2-player design improves engagement
- Winner display with animation and performance metrics

Further Discussion of Results

- Add multiple difficulty modes (beginner to advanced)
- Improve enclosure design (compact, portable device)
- Introduce single-player training mode
- Store and analyze player statistics (reaction time trends)
- Enhance visuals and audio for better user experience

Thank You

